

Coordination Compounds

NCERT Class XII Chemistry · Chapter 5 · Coordination Compounds

Coordination Compounds — Complete Notes

11 marks, and entirely new ground. This chapter has more vocabulary than any other in the paper, so nothing here assumes you've met a term before — every one tagged [exposure] is defined in plain words, with a worked example, before it gets used again.

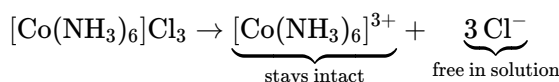
What a coordination compound actually is — *start here — the whole chapter builds on this one picture*

[exposure] A **coordination compound** is a compound in which a central metal atom or ion is surrounded by, and bonded to, a fixed number of molecules or negative ions. Transition metals form these in huge numbers because they have small, highly charged ions and empty d orbitals ready to accept electron pairs. The bond isn't the usual "one electron each" kind — the surrounding species donates *both* electrons. A standard example is $[\text{Co}(\text{NH}_3)_6]^{3+}$: one cobalt ion at the centre, six ammonia molecules attached around it.

They matter well beyond the exam: **chlorophyll** is a coordination compound of magnesium, **haemoglobin** of iron, and **vitamin B₁₂** of cobalt. Industrially they're used in electroplating, dyeing, medicine, metal extraction and catalysis.

The square bracket is the whole notation

Everything **inside** the square bracket travels together as one unit and does *not* break apart in water. Everything **outside** is a separate ion that dissociates freely.



Complex vs double salt — the distinction they examine

	Double salt	Complex
In water	Dissociates completely into all its ions	The bracketed part keeps its identity
Example	Mohr's salt $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6 \text{H}_2\text{O}$	$\text{K}_4[\text{Fe}(\text{CN})_6]$
What you detect	Fe^{2+} gives a positive test	No free Fe^{2+} — the complex ion survives

Also double salts: carnallite $\text{KCl} \cdot \text{MgCl}_2 \cdot 6 \text{H}_2\text{O}$ and potash alum $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$. They exist only in the solid state.

Examiner asks: "difference between a complex and a double salt" (2019) — answer with the water behaviour and one example each.

Werner's theory — *the 1898 idea that made the whole subject make sense*

Alfred Werner explained a puzzle: why does CoCl_3 combine with different numbers of NH_3 to give differently coloured compounds with different behaviour? His answer was that a metal has **two kinds of valence**.

[exposure] Primary valence is the metal's ordinary oxidation number — the charge it carries. It is **ionisable**, meaning whatever satisfies it can break away in water, and it is satisfied by **negative ions**. In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ cobalt is +3, so its primary valence is 3, satisfied by the three chloride ions sitting outside the bracket — which is exactly why those three come free in solution.

[exposure] Secondary valence is the number of species directly bonded to the metal — what we now call the coordination number. It is **non-ionisable**, so these stay attached in water, and it's satisfied by neutral molecules or negative ions that donate a lone pair. It is **fixed for a given metal**. In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ the secondary valence is 6, satisfied by the six ammonia molecules inside the bracket. Those ligands arrange themselves in a definite shape in space — the **coordination polyhedron**.

How Werner proved it — the silver nitrate experiment

Add excess AgNO_3 to each compound. Only **free** chloride ions precipitate as AgCl , so counting the moles of AgCl tells you how many chlorides are outside the bracket:

Formula	Colour	Moles AgCl	Actual structure
$\text{CoCl}_3 \cdot 6 \text{NH}_3$	Yellow	3	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
$\text{CoCl}_3 \cdot 5 \text{NH}_3$	Purple	2	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
$\text{CoCl}_3 \cdot 4 \text{NH}_3$	Green	1	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

Notice the coordination number stays **6** throughout — as ammonia leaves the bracket, chloride moves in to take its place.

The vocabulary — *nine terms, each with its own example*

[exposure] Central atom or ion — the metal at the middle, the one everything else attaches to. It acts as a **Lewis acid**: it's electron-deficient, so it *accepts* electron pairs. In $[\text{Ni}(\text{H}_2\text{O})_4\text{Cl}_2]$ the central ion is Ni^{2+} ; in $[\text{Fe}(\text{CN})_6]^{3-}$ it's Fe^{3+} .

[exposure] Ligand — a molecule or ion attached to the central metal. It must have at least one **lone pair of electrons** to donate, which is what lets it act as a **Lewis base**. Ligands range from simple ions (Cl^- , CN^-) through small molecules (H_2O , NH_3) to whole proteins. The specific atom in the ligand that donates the pair is the **donor atom** — in NH_3 it's the nitrogen.

[exposure] Coordinate bond — the bond formed when *one* species supplies *both* electrons of the shared pair. It exists as a separate name because an ordinary covalent bond takes one electron from each partner. Once formed it's indistinguishable from a normal covalent bond; only its origin differs. Every metal-ligand bond in this chapter is one.

[exposure] Coordination number — the number of donor atoms directly bonded to the metal. Count **donor atoms, not ligands**, and count only sigma bonds. In $[\text{Co}(\text{NH}_3)_6]^{3+}$ six

ammonias give six donor atoms, so CN = 6. But in $[\text{Co}(\text{en})_3]^{3+}$ there are only three ligands — each supplies *two* nitrogen donors, so $\text{CN} = 3 \times 2 = 6$ again.

[exposure] Coordination sphere — the central metal plus its ligands, written inside the square bracket together with the overall charge. Whatever sits outside is the **counter ion**, which is the ionisable part. In $\text{K}_4[\text{Fe}(\text{CN})_6]$ the coordination sphere is $[\text{Fe}(\text{CN})_6]^{4-}$ and the counter ions are four K^+ .

[exposure] Coordination polyhedron — the shape the ligands make in space around the metal. It exists as a term because the geometry, not just the count, determines the compound's properties and isomers. Six ligands give an **octahedron** (90° angles), four give either a **tetrahedron** (109.5°) or a **square planar** arrangement (90° , all in one plane).

Oxidation number of the central atom

The charge the metal would carry if every ligand were removed along with its donated electron pair. Write it as a Roman numeral in brackets after the metal's name.

Method: set the metal as x , add each ligand's charge, set the sum equal to the overall charge on the bracket. For $[\text{Cu}(\text{CN})_4]^{3-}$: $x + 4(-1) = -3$, so $x = +1 \rightarrow$ copper(I).

Homoleptic vs heteroleptic

Homoleptic — one type of ligand only, e.g. $[\text{Co}(\text{NH}_3)_6]^{3+}$. **Heteroleptic** — more than one type, e.g. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

Types of ligand — *denticity, ambidentate, and the chelate effect*

[exposure] Denticity is simply how many donor atoms a single ligand uses to grip the metal — from Latin *dens*, tooth. The word exists because a ligand with several donor atoms behaves very differently from one with a single donor: it can clamp onto the metal at multiple points at once. A ligand with one donor site is **unidentate**, two is **didentate**, three or more is **polydentate**.

Type	Donor sites	Examples
Unidentate	1	Cl^- , NH_3 , H_2O , OH^- , CN^-
Didentate	2	ethane-1,2-diamine (en); oxalate $\text{C}_2\text{O}_4^{2-}$ (ox)
Polydentate	3+	EDTA^{4-} — hexadentate

EDTA^{4-} is worth knowing precisely: it binds through **two nitrogen atoms and four oxygen atoms**, six donor sites in total.

[exposure] An ambidentate ligand is a unidentate ligand that has *more than one kind* of donor atom available, and can attach through either one — but only one at a time. It's a distinct category because it creates isomers that differ purely in which atom did the bonding. NO_2^- can bind through nitrogen (called **nitrito-N**) or through oxygen (**nitrito-O**). SCN^- can bind through sulphur (**thiocyanato**) or nitrogen (**isothiocyanato**).

[exposure] A chelate ligand is a didentate or polydentate ligand that grips one metal ion at two or more points, forming a **ring** that includes the metal — the word comes from the Greek

for a crab's claw. It matters because chelate complexes are markedly **more stable** than equivalent complexes made from separate unidentate ligands. In $[\text{Pt}(\text{en})\text{Cl}_2]$, the "en" ligand bonds through both its nitrogens, closing a five-membered ring around the platinum.

Why chelates are more stable: replacing six separate unidentate ligands with three didentate ones releases more free particles into solution, so entropy rises; combined with a favourable enthalpy this makes ΔG more negative. More rings means more stability.

IUPAC nomenclature — *the rules in order, then worked examples*

Naming a complex — the rules

1. **Cation first, then anion** — as in any ionic compound.
2. **Ligands before the metal**, in **alphabetical order** regardless of charge.
3. **Anionic ligands end in -o**: chlorido, bromido, cyanido, hydroxido, sulphato, oxalato.
4. **Neutral ligands** keep special names: H_2O = **aqua**, NH_3 = **ammine** (two m's), CO = **carbonyl**, NO = nitrosyl.
5. **Prefixes** di, tri, tetra for simple ligands; **bis, tris, tetrakis** when the ligand's own name already contains a prefix (then bracket the ligand).
6. Prefixes are **ignored** when alphabetising.
7. **Oxidation number** in Roman numerals, in brackets, right after the metal.
8. If the complex ion is an **anion**, the metal name ends in **-ate** (ferrate, cuprate, argentate, zincate, cobaltate).

Worked — three names built step by step

$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$: alphabetise ammine, aqua, chlorido. Four ammines → tetraammine.
Charge: $x + 0 + 0 - 1 = +2$, so $x = +3$.

→ **tetraamminaquachloridocobalt(III) chloride**

$[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$: neutral overall, so $x - 3 = 0$, $x = +3$.

→ **triamminetrichloridochromium(III)**

$[\text{PtCl}_2(\text{en})_2]^{2+}$: "c" before "e", so chlorido first. "en" already contains "di", so use **bis** and bracket it. $x + 0 - 2 = +2$, $x = +4$.

→ **dichloridobis(ethane-1,2-diamine)platinum(IV)**

$\text{K}_4[\text{Fe}(\text{CN})_6]$: anionic complex, so iron becomes **ferrate**. $x - 6 = -4$, $x = +2$.

→ **potassium hexacyanidoferrate(II)**

Writing the formula from a name

1. Metal first, then ligands in **alphabetical order**.
2. Polydentate and abbreviated ligands go in **parentheses**; alphabetise abbreviations by their first letter (en → e, ox → o).
3. Whole coordination entity in **square brackets**, no spaces inside.
4. Charge as a right superscript outside the bracket, number before sign.

Trap: "ammine" with two m's is NH_3 ; "amine" with one m is an organic amine like ethane-1,2-diamine. NCERT uses both, in these exact senses. Also: current IUPAC is **chlorido**, not chloro.

Isomerism — two families, seven types — the biggest mark-earner in the chapter

[exposure] Isomers are compounds with the same formula but a different arrangement of atoms. In coordination chemistry they split into two families: **stereoisomers**, where the connections are the same but the spatial arrangement differs, and **structural isomers**, where what is bonded to what actually differs. The distinction matters because the two families are detected in completely different ways — stereoisomers by geometry and optical activity, structural ones by simple chemical tests.

Stereoisomerism 1 • Geometrical (cis-trans)

Same ligands, different positions. **Cis** = identical ligands adjacent; **trans** = opposite.

- **Square planar** MX_2L_2 — shows cis and trans. Example: $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.
- **Octahedral** MX_2L_4 — shows cis and trans. Example: $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.
- **Octahedral** $[\text{CoCl}_2(\text{en})_2]^+$ — cis and trans.
- **Octahedral** MA_3B_3 — gives **facial (fac)** where the three identical ligands occupy one triangular face, and **meridional (mer)** where they lie around a meridian. Example: $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$.

Trap: tetrahedral complexes do not show geometrical isomerism. In a tetrahedron every position is adjacent to every other — all angles stay 109.5° however you swap ligands, so there's no "opposite" to be trans to. This is a standing exam question.

Stereoisomerism 2 • Optical

[exposure] A complex is **optically active** when it has no plane of symmetry, so its mirror image cannot be superimposed on it — like your left and right hands. Such a molecule is **chiral**, and the two mirror-image forms are **enantiomers**: one rotates plane-polarised light to the right (**dextro, d**), the other equally to the left (**laevo, l**). It matters because two compounds identical in every other property can be told apart this way.

Most common in octahedral complexes with didentate ligands:

- $[\text{Cr}(\text{ox})_3]^{3-}$ — type $\text{M}(\text{AA})_3$, optically active.
- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ — type $\text{M}(\text{AA})_2\text{X}_2$: the **cis** form is optically active, the **trans** form is not (its mirror image is superimposable).

Examiner asks: "draw the geometrical isomer of $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$ which is optically active" — draw the **cis** form. Trans is always the optically inactive one in these.

Structural isomerism — four types

Type	What differs	Example pair
Linkage	Which atom of an ambidentate ligand bonds	$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ vs $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$
Ionisation	Ligand and counter ion swap places	$[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ vs $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$
Hydrate	Water inside vs outside the sphere	$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet) vs $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
Coordination	Ligands swap between a cationic and anionic complex	$[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ vs $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$

The chemical test for ionisation isomers (asked directly): add BaCl_2 — the isomer with free sulphate gives a white BaSO_4 precipitate. Add AgNO_3 — the isomer with free chloride gives white AgCl . Whichever ion is *outside* the bracket is the one that reacts.

Valence bond theory — the hybridisation method — work every NCERT example the same way

VTB says the metal makes empty **hybrid orbitals** of equal energy, and each ligand donates a lone pair into one of them.

CN	Hybridisation	Geometry
4	sp^3	Tetrahedral
4	dsp^2	Square planar
6	d^2sp^3	Octahedral (inner orbital)
6	sp^3d^2	Octahedral (outer orbital)

[exposure] A **strong field ligand** pushes hard enough on the metal's d electrons to force them to **pair up**, freeing inner d orbitals for bonding. A **weak field ligand** doesn't, so the electrons stay unpaired. This single distinction decides the hybridisation, the geometry, and the magnetism of every complex below. **Strong:** CN^- , CO, and NH_3 when the metal is highly charged (Co^{3+} , Cr^{3+}). **Weak:** F^- , Cl^- , H_2O .

The method — five steps, every time

1. Find the metal's oxidation number, then its d-electron count.
2. Draw the d orbitals and fill them.
3. Ask: strong ligand (pair up) or weak (leave unpaired)?
4. Count how many orbitals the ligands need, and take that many empty ones — **inner** 3d if pairing freed some, otherwise **outer** 4d.
5. Name the hybridisation, the geometry, and the magnetism.

Complex	Metal	Ligand	Hybrid	Geometry	Magnetism
$[\text{Co}(\text{NH}_3)_6]^{3+}$	$\text{Co}^{3+}, 3d^6$	Strong	d^2sp^3	Octahedral	Diamagnetic
$[\text{CoF}_6]^{3-}$	$\text{Co}^{3+}, 3d^6$	Weak	sp^3d^2	Octahedral	Paramagnetic
$[\text{NiCl}_4]^{2-}$	$\text{Ni}^{2+}, 3d^8$	Weak	sp^3	Tetrahedral	Paramagnetic
$[\text{Ni}(\text{CN})_4]^{2-}$	$\text{Ni}^{2+}, 3d^8$	Strong	dsp^2	Square planar	Diamagnetic
$[\text{Ni}(\text{CO})_4]$	$\text{Ni}, 3d^8 4s^2$	Strong	sp^3	Tetrahedral	Diamagnetic

Worked — $[\text{Ni}(\text{CO})_4]$, the one that catches people out

CO is neutral, so nickel's oxidation state is **0** — configuration $3d^8 4s^2$.

CO is a strong ligand: it forces pairing. The two 4s electrons move into 3d, filling it to $3d^{10}$ and leaving 4s empty.

Four CO ligands need four orbitals: one 4s + three 4p $\rightarrow sp^3$, tetrahedral. All electrons paired $\rightarrow$ **diamagnetic**.

Trap: $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ have the same metal in the same oxidation state but *different* geometry and magnetism — purely because Cl^- is weak and CN^- is strong. This exact pair is a favourite comparison.

[exposure] Inner orbital (or **low spin**, or **spin-paired**) complexes use the metal's *inner* 3d orbitals, which only become free when a strong ligand forces pairing — so they have few or no unpaired electrons. **Outer orbital (high spin, spin-free)** complexes use the outer 4d orbitals because a weak ligand left the 3d electrons unpaired and in the way. The names are just describing which shell supplied the orbitals: $[\text{Co}(\text{NH}_3)_6]^{3+}$ is inner/low-spin, $[\text{CoF}_6]^{3-}$ is outer/high-spin.

Magnetic moment

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

Measuring μ tells you n , which tells you whether the ligand was strong or weak — this is how VBT is tested experimentally.

Limitations of VBT — asked as a list

1. Involves many assumptions.
2. Gives no quantitative account of magnetic data.
3. Does not explain the **colour** of complexes.
4. Gives no quantitative account of stability.
5. Cannot predict whether a 4-coordinate complex will be tetrahedral or square planar.
6. Does not distinguish weak from strong ligands — it just assumes the answer.

Crystal field theory — *the model that finally explains colour*

[exposure] Crystal field theory treats the metal-ligand bond as purely **electrostatic attraction** — the positive metal ion pulling on negative ions or on the negative end of polar

molecules. It replaced VBT's covalent picture because it can explain what VBT could not: why complexes are coloured, and why some are high spin and others low. Anionic ligands like Cl^- are treated as point negative charges; neutral ones like H_2O as dipoles pointing their negative end at the metal.

[exposure] Crystal field splitting is the central idea. In a free metal ion all five d orbitals have identical energy — they are **degenerate**. If you surrounded the ion with an evenly spread sphere of negative charge, all five would rise in energy equally and stay degenerate. But real ligands arrive from *specific directions*, so some d orbitals get pushed much harder than others. The degeneracy breaks and the five orbitals split into two groups at different energies. That energy gap is what visible light interacts with — which is where colour comes from.

Octahedral field

Ligands approach **along the x, y and z axes**.

- $d_{x^2-y^2}$ and d_{z^2} point **along** the axes → head-on repulsion → energy **raised**. These two are the e_g set.
- d_{xy} , d_{yz} , d_{zx} point **between** the axes → less repulsion → energy **lowered**. These three are the t_{2g} set.

The gap is the **crystal field splitting energy**, Δ_o .

High spin or low spin — the Δ_o vs P contest

[exposure] Pairing energy (P) is the energy cost of forcing a second electron into an orbital that already has one, against their mutual repulsion. It matters because the fourth d electron faces a genuine choice: pay P and pair up in the lower t_{2g} set, or pay Δ_o and go alone into the upper e_g set. Whichever costs less, wins.

- $\Delta_o < P$ (weak field ligand): cheaper to jump up. Fourth electron enters $e_g \rightarrow t_{2g}^3 e_g^1 \rightarrow$ more unpaired electrons → **high spin**.
- $\Delta_o > P$ (strong field ligand): cheaper to pair. Fourth electron pairs in $t_{2g} \rightarrow t_{2g}^4 \rightarrow$ fewer unpaired → **low spin**.

Tetrahedral field

The geometry inverts: now d_{xy} , d_{yz} , d_{zx} lie closer to the incoming ligands and are **raised** (t_2), while $d_{x^2-y^2}$ and d_{z^2} are **lowered** (e).

The splitting is much smaller, because there are only four ligands instead of six and they don't point directly at any orbital:

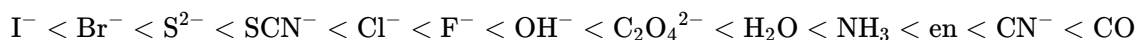
$$\Delta_t = \frac{4}{9}\Delta_o$$

Since Δ_t is always smaller than P, **tetrahedral complexes are essentially always high spin**.

Examiner asks: "how is Δ_o related to Δ_t ?" (2024) — quote $\Delta_t = \frac{4}{9}\Delta_o$ and say why: fewer ligands, none pointing directly at an orbital.

Spectrochemical series

Ligands arranged by the size of Δ they produce, determined experimentally from light absorption. Weak field on the left, strong on the right:



Colour, explained at last

An electron absorbs light of exactly the energy Δ_o and jumps from t_{2g} to e_g — a **d-d transition**. The compound transmits what it didn't absorb, so we see the **complementary** colour. A complex with d^0 or d^{10} has no possible jump and is colourless.

Examiner asks: why does a complex lose its colour when the ligands are removed? Without ligands there is no crystal field, so no splitting, so no d-d transition — it turns colourless.

Stability and applications — *the short sections that still carry marks*

Stability constant

For $\text{M} + 4\text{L} \rightleftharpoons \text{ML}_4$, the stability (formation) constant is $K = \frac{[\text{ML}_4]}{[\text{M}][\text{L}]^4}$. Higher K means a more stable complex. Stability rises with the metal's charge, with smaller metal size, and — strongly — with chelation.

Applications worth naming

- **Wilkinson's catalyst**, $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ — catalyses the **hydrogenation of alkenes**.
- **Cisplatin**, $\text{cis-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ — anti-cancer drug. Only the *cis* isomer works.
- **EDTA** — estimation of water hardness; treatment of lead poisoning.
- **Extraction of metals** — gold and silver via cyanide complexes.
- **Biological** — chlorophyll (Mg), haemoglobin (Fe), vitamin B₁₂ (Co).
- **Electroplating** and photography (silver thiosulphate complex).

Patterns, collected — *the four things you'll be asked to actually do*

A · IUPAC name $\leftrightarrow$ formula — 2-3 marks

Recognise it: "write the IUPAC name" or "write the formula".

1. Get the oxidation number first — it's needed either way.
2. Naming: ligands alphabetically, anionic ones in -o, prefixes ignored for alphabetising, Roman numeral, -ate if the complex is an anion.
3. Formula: metal first, then ligands alphabetically, abbreviations in parentheses.

B · Hybridisation, geometry, magnetism — 3 marks

Recognise it: an atomic number is given alongside a complex.

1. Oxidation number $\rightarrow$ d-electron count.

2. Strong or weak ligand? Pair or don't.
3. Count the orbitals needed; inner 3d if free, else outer 4d.
4. State hybridisation, geometry, and paramagnetic/diamagnetic.

C · Identify the isomerism — 2-3 marks

Recognise it: "what type of isomerism is shown by..."

1. Ambidentate ligand (NO_2^- , SCN^-) present → **linkage**.
2. An ion could swap in/out of the bracket → **ionisation**.
3. Water present both inside and out → **hydrate**.
4. Two complex ions in one compound → **coordination**.
5. Otherwise check geometry: square planar or octahedral → **geometrical**; didentate ligands → check **optical** too.

D · Magnetic moment — 2 marks

Recognise it: "calculate the magnetic moment" or "predict the magnetic behaviour".

1. d-count, then apply the ligand's field strength.
2. Count unpaired n .
3. $\mu = \sqrt{n(n+2)}$ BM.

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · IUPAC naming / formula writing — 2-3 marks

Recognise it: a complex given with "name this" or a name given with "write the formula".

1. Oxidation number.
2. Alphabetical ligands, correct endings.
3. -ate if anionic.

Trap: "ammine" vs "amine", and forgetting **bis/tris** when the ligand name already has a prefix.

2 · Isomerism identification and drawing — 2-3 marks

Recognise it: "what type of isomerism", or "draw the isomer which is optically active".

1. Scan for ambidentate ligands, swappable counter ions, or two complex ions.
2. If none, it's stereoisomerism — check cis/trans, then optical.
3. Optically active = **cis** with didentate ligands.

3 · Hybridisation and magnetic behaviour — 3 marks

Recognise it: atomic number supplied.

1. Oxidation number → d count.
2. Ligand strength → pairing.
3. Hybridisation → geometry → magnetism.

Trap: in $[\text{Ni}(\text{CO})_4]$ nickel is in oxidation state **zero**, not +2.

4 · Crystal field theory reasoning — 3 marks

Recognise it: Δ_o , Δ_t , high/low spin, or "why is it coloured".

1. Compare Δ_o with P.
2. $\Delta_o < P \rightarrow$ high spin; $\Delta_o > P \rightarrow$ low spin.
3. For colour, name the d-d transition.

5 · Definitions and comparisons — 2 marks

Recognise it: "define", "difference between", "what is denticity/chelate/ambidentate".

1. Give the definition in NCERT's own words.
2. Add one concrete example — it's usually worth half the mark.

6 · Chemical tests and applications — 2-3 marks

Recognise it: "give a chemical test", or "write the formula and use of..."

1. Ionisation isomers $\rightarrow$ BaCl_2 for sulphate, AgNO_3 for chloride.
2. Named catalysts: Wilkinson's for alkene hydrogenation.

Past year questions · mark slots — what each type is worth

Question type	Slot	Time
IUPAC naming / formula	2-3 marker	3 min
Isomerism ID + drawing	2-3 marker	3-4 min
Hybridisation + magnetism	3-marker	4 min
CFT reasoning	3-marker; full splitting diagram is a 5	4-6 min
Definitions / differences	2-marker	2 min
Chemical test / application	2-marker	2 min
Multi-part "answer the following"	5-marker	7 min

The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — the multi-part sets combining naming, isomerism and CFT are where the 5-markers sit.

Past year questions · repeat offenders — appeared more than once — highest probability in the chapter

2026 Q1 · 2019 Q4 · 2023 Q2

IUPAC naming — including naming the **linkage isomer** of a red complex, and writing a formula from a given name. Naming appears in essentially every paper.

2017 Q14 · 2022 Q19 · 2026 Q5

"What type of isomerism is shown by..." — run across linkage, ionisation and coordination isomers in the same question format.

2025 Q13 · 2022 Q16

Hybridisation and magnetic behaviour of $[\text{Ni}(\text{CO})_4]$ with $Z = 28$ supplied. The oxidation-state-zero trap is the point of it.

2024 Q24(a) · 2026 Q17

$\Delta_t = \frac{4}{9}\Delta_o$, and the Δ_o vs pairing-energy comparison deciding high vs low spin.

2019 Q25(a) · lecture PYQ

Difference between a complex and a double salt, with examples. Straight recall, always available.

2025 Q18 · 2022 Q23

Geometrical isomerism: which complexes show it, fac vs mer for $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, and why tetrahedral complexes don't.

2025 Q22 · 2026 Q20

Wilkinson's catalyst formula and use, and why a complex becomes colourless when ligands are absent.

Past year questions · cold practice — *answers only* — *work them before you look*

Naming and formulae

- 2019 — write the formula for pentaamminenitrito-O-cobalt(III). $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$
- Lecture PYQ — IUPAC name of $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$. triamminetrichloridochromium(III)
- Lecture PYQ — IUPAC name of $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$. pentaamminecarbonatocobalt(III) chloride
- Lecture PYQ — IUPAC name of $[\text{Zn}(\text{OH})_4]^{2-}$. tetrahydroxidozincate(II) ion

Isomerism

- 2026 Q5 — isomerism in $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$. Linkage — NO_2^- is ambidentate
- 2017 Q14 — isomerism in $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. Ionisation — sulphate and chloride can swap
- 2017 Q14 — isomerism in $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$. Coordination — ligands exchange between the two spheres
- 2025 Q18 — draw the optically active geometrical isomer of $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$. The **cis** form; trans is optically inactive
- Lecture PYQ — which shows geometrical isomerism: $[\text{Co}(\text{NH}_3)_6]^{3+}$ or $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$? The second — fac and mer. The first has all identical ligands

Bonding

- 2025 Q13 — hybridisation and magnetic behaviour of $[\text{Ni}(\text{CO})_4]$, $Z = 28$. sp^3 , tetrahedral, diamagnetic — Ni is in oxidation state 0
- Lecture PYQ — compare $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$. sp^3 tetrahedral paramagnetic vs dsp^2 square planar diamagnetic

- Lecture PYQ — why is $[\text{CoF}_6]^{3-}$ paramagnetic but $[\text{Co}(\text{NH}_3)_6]^{3+}$ diamagnetic? F^- weak (no pairing, sp^3d^2); NH_3 strong with Co^{3+} (pairs, d^2sp^3)

CFT and applications

- 2024 Q24(a) — relate Δ_o and Δ_t . $\Delta_t = \frac{4}{9}\Delta_o$ — fewer ligands, none pointing directly at an orbital
- 2026 Q17 — what does $\Delta_o < P$ mean for the complex? Weak field, high spin — the fourth electron goes to e_g rather than pairing
- 2025 Q22 — formula and use of Wilkinson's catalyst. $[(\text{Ph}_3\text{P})_3\text{RhCl}]$; hydrogenation of alkenes
- 2026 Q6 — chemical test distinguishing $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ from $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. $\text{AgNO}_3 \rightarrow$ white AgCl for the first; $\text{BaCl}_2 \rightarrow$ white BaSO_4 for the second

Built from Sourabh Raina's Coordination Compounds one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 5 (Rationalised 2022–23). Nomenclature verified against NCERT: **ammine** (two m's) for NH_3 versus **amine** for organic amines, and **chlorido** rather than chloro. First-contact material: every term marked **[exposure]** is explained from scratch before use.